

Different signals, identical samples

Background

Sampling can hide differences between continuous-time signals. For integer sample indices, zero-phase cosines whose frequencies differ by an integer multiple of f_s have identical samples. This is an example of aliasing ambiguity.

Your task

Sample two unit-amplitude, zero-phase cosines at the same N instants. Return their sequences and the largest absolute difference. Decide whether the difference is at most $1e-10$. Use the supplied plot to compare continuous curves and sampled values.

Function contract and assumptions

`compare_cosines(f1, f2, fs, N)` returns `(t, x1, x2, error, same)`. The arrays have length N and t starts at zero. `error` is $\max |x1[n]-x2[n]|$; `same` is a Boolean. Frequencies are nonnegative, $f_s > 0$, and $N \geq 1$.

Example / correctness check

$f_1=3$, $f_2=23$, $f_s=20$, $N=20$ should give `error` approximately zero and `same=True`. Replacing f_2 with 7 should give `same=False`.

Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 5-6 and 22.